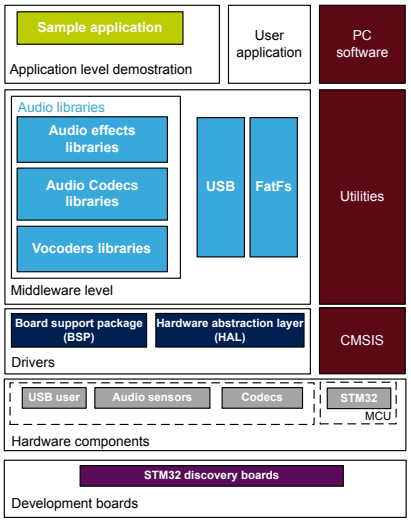
X-CUBE-AUDIO

Data brief

Audio effects libraries software expansion for STM32Cube



Product status link

X-CUBE-AUDIO

Features

- STM32Cube audio effect modules and libraries for the STM32F4 and STM32F7 Series of microcontrollers
- 16-bit wave format stereo input at 48 kHz expected by Xcube audio applications
- 32-bit resolution audio library computation and 16- or 32-bit I/O buffers supported
- Bass manager (BAM): management of the low frequencies of a stereo signal, including compressor and limiter
- Generic biquad filter (BIQ): IIR second order filters and predefined standard filters such as peak removal, loudness, notch, voice enhancer, low-pass, and high-pass frequency response for transducer equalization
- Clock drift compensation (CDC): compensation of drift by smoothly adding or removing one sample
- Gain manager (GAM): management of input signal volume with negative gains in range [-80 dB: 0 dB] with 0.5 dB granularity without compression
- Graphical equalizer (GREQ): 5, 8 or 10 bands. Adjustable gain factors from -12 dB to +12 dB in standard mode
- Headphone virtualizer (HPV): audio virtualization for headphones from a 1.0/2.0/5.1/7.1 input stream to a virtualized 2.0 output stream
- MP3 Decoder (MP3Dec): decoder for MPEG-1,2 or 2.5 formats (for layers 1, 2 and 3) supporting constant, free format or VBR variable bit rate for mono or stereo audio input streams with PCM (Pulse Code Modulation) decoded output.
- MP3 encoder (MP3Enc): encoder for MPEG-1,2 or 2.5 formats (for layer 3 only) supporting fixed or free format bit rate for mono or stereo audio input streams.
- Sampling rate converter (SRC236 and SRC441):
  - sampling frequency conversion from any rate with a ratio of 2, 3, 6, 3/2, 1/2, 1/3, 1/6, or 2/3
  - sampling frequency conversion from 44.1 kHz to 48 kHz
- Omni surround multichannel virtualizer (OMNI2):
  - audio virtualization for loudspeakers from a 1.0/2.0/5.1/7.1 input stream to a virtualized 2.0 output stream (including stereo widening effect).
  - Omni surround stereo widener (OMNI2_SW only) audio virtualization for loudspeakers from a 1.0/2.0 input stream to a widened 2.0 output stream.
- Smart volume control (SVC): management of audio input signal volume including a compression with gains in the range [-80 dB:+36 dB]
- Sound detector (SDR): audio signals detection used to trigger signal processing such as speech recognition
- Sound meter (SMR): level measurement on a logarithmic scale

Description

The X-CUBE-AUDIO package contains a comprehensive set of audio processing components for the STM32F4 and STM32F7 Series of microcontrollers. It is composed of high-quality efficient software libraries and modules ready to be embedded into a variety of audio appliances. All audio binaries are delivered with internal 32-bit processing and support either 16- or 32-bits I/O buffers.

Each audio effect application in X-CUBE-AUDIO expects a 16-bit stereo audio input signal at a 48-kHz sampling rate, using wave format with I/O data buffers dimensioned for 10 ms.

General information

The X-CUBE-AUDIO embedded software package runs on STM32 32-bit microcontrollers based on the Arm® Cortex®-M processor.

Note: Arm is a registered trademark of Arm Limited (or its subsidiaries) in the US and/or elsewhere.

Ordering Information

X-CUBE-AUDIO is available for free download from the www.st.com website.

License

X-CUBE-AUDIO is delivered under the *Mix Liberty + OSS + 3rd party V1* license.

The software components provided in this package come with different license schemes as shown in Table 1. Software component license agreements.

For more details, refer to the license agreement of each component.

Table 1. Software component license agreements

Software component	Owner	License
Board Support Package (BSP)	ST	Open source BSD
Cortex®-M CMSIS	Arm®	Open source BSD
HAL STM32 F4/F7	ST	Open source BSD
Libraries for audio effects modules (as PDM, BAM, BIQ, CDC,GAM,GREQ, HPV, OMNI2, OMNI2_SW, SDR, SMR, SRC236, SRC441, SVC)	ST	Image V2
Libraries for Audio codecs: MP3 decoder	ST	Image V2
Libraries for Audio codecs: spiritDSP MP3 encoder	Spirit DSP	Image V2 (binary release)
Libraries for Audio codecs: spiritDSP MP3 decoder	Spirit DSP	Image V2 (binary release)
Libraries for Audio codecs: spiritDSP MP3 wrapper	ST	Ultimate Liberty (source release)
Project examples & Common files	ST	Ultimate Liberty (source release)

Revision history

Table 2. Document revision history

Date	Revision	Changes
03-Jun-2016	1	Initial release.
01-Feb-2018	2	Updated Section Features, Section Description. Replaced X-CUBE-AUDIO-F4 and X-CUBE-AUDIO-F7 RPNs with X-CUBE-AUDIO. Added Section License
03-Jul-2018	3	Updated Section Features and Table 1. Software component license agreements
19-Sep-2018	4	Updated Section Features, Table 1. Software component license agreements
05-Nov-2018	5	Updated Section Features and Table 1. Software component license agreements
03-Apr-2019	6	Updated image on front page, Section Features, Section Description, Table 1. Software component license agreements.

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